

Lesson 6: Logarithms & Their Laws — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Evaluate $\log_2 16$.
2. Evaluate $\log_5 25$.
3. Evaluate $\log_{10} 10000$.
4. Rewrite $2^6 = 64$ as a logarithm.
5. Evaluate $\log_3 1$.
6. Evaluate $\log_7 7$.

Section 2 — Core practice

7. Write as a single logarithm: $\log 6 + \log 4$.
8. Write as a single logarithm: $\log 40 - \log 8$.
9. Write $3 \log x$ as a single logarithm.
10. Simplify $\log_2 8 + \log_2 4$.
11. Write $2 \log a + \log b$ as a single logarithm.

12. Evaluate $\log_4 64$.

Section 3 — Challenge & exam-style

13. Given $\log_2 3 = k$, write $\log_2 24$ in terms of k .

14. Simplify $\log x^3 - \log x$.

15. Solve $\log_3(x) = 4$.

16. Given $\log a = 0.7$ and $\log b = 0.3$, find $\log(a^2/b)$.